The Husimi Function of the Kicked Top

Spin coherent states, the phase-space picture of a quantum spin and of a qubit chain, and how the `classical_kicked_top` code computes it

Abstract

The Husimi function $Q(\theta,\phi)$ turns a quantum spin state ($2j+1$ complex amplitudes) into a picture on the classical phase-space sphere, so quantum and classical kicked-top dynamics can be compared on one plot. We build Q from scratch for a single spin and for a chain of n_q qubits, link each result to the Fortran routine that implements it, and write out all algebra.

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Notation

We use units with $\hbar = 1$. The polar angle $\theta \in [0,\pi]$ is measured from the z axis, the azimuth $\phi \in [-\pi,\pi)$ around it, and

$$\hat{\boldsymbol{n}}(\theta,\phi) = \boldsymbol{s} = (X,Y,Z) = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta), \qquad Z = \cos\theta, \tag{1}$$

is the unit vector in that direction. A hat on a bold symbol always means a unit vector: $\hat{\boldsymbol{x}} = (1,0,0)$, $\hat{\boldsymbol{y}} = (0,1,0)$, $\hat{\boldsymbol{z}} = (0,0,1)$, and $\hat{\boldsymbol{n}}, \hat{\boldsymbol{n}}', \hat{\boldsymbol{n}}_0$ are directions on the sphere. Quantum operators carry *no* hat. The spin operators $\boldsymbol{J} = (J_x, J_y, J_z)$ satisfy

$$[J_a, J_b] = i\,\epsilon_{abc}\,J_c, \tag{2}$$

where $[A,B] = AB - BA$, i is the imaginary unit, $a,b,c \in \{x,y,z\}$, c is summed over, and ϵ_{abc} is the Levi-Civita symbol ($\epsilon_{xyz} = \epsilon_{yzx} = \epsilon_{zxy} = 1$, $\epsilon_{xzy} = \epsilon_{zyx} = \epsilon_{yxz} = -1$, zero if two indices are equal). For spin j ($j = \frac{1}{2}, 1, \frac{3}{2}, \dots$) the basis states $|j,m\rangle$, $m = j, j-1, \dots, -j$, satisfy $\boldsymbol{J}^2|j,m\rangle = j(j+1)|j,m\rangle$, $J_z|j,m\rangle = m|j,m\rangle$, and with the ladder operators $J_\pm = J_x \pm iJ_y$

$$J_+|j,m\rangle = a(m)\,|j,m+1\rangle, \quad J_-|j,m\rangle = a(m-1)\,|j,m-1\rangle, \quad a(m) = \sqrt{j(j+1) - m(m+1)}. \tag{3}$$

A state is $|\psi\rangle = \sum_m \psi_m |j,m\rangle$. In the code (`spin_mod`) ψ_m is stored at array index $i = 1, \ldots, 2j+1$ with $m = j - (i-1)$. $\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$ is the binomial coefficient. The kicked top has the Hamiltonian and the one-period (Floquet) operator, sampled just before each kick,

$$H(t) = \frac{k}{2j} J_z^2 + p\, J_y \sum_{n=-\infty}^{\infty} \delta(t-n), \qquad U = e^{-i\frac{k}{2j}J_z^2}\, e^{-ipJ_y}, \tag{4}$$

with torsion strength k and kick angle p; $|\psi_n\rangle = U^n|\psi_0\rangle$ is the state after n kicks. The rotation matrices about z and y (counter-clockwise by the right-hand rule) are

$$R_z(\alpha) = \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad R_y(\alpha) = \begin{pmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{pmatrix}. \tag{5}$$

1 Motivation: a phase-space picture for a quantum spin

The classical phase space is a sphere. Classically, the kicked top is a vector $\boldsymbol{J}$ of fixed length j; only its direction $\boldsymbol{s} = \boldsymbol{J}/j$ changes. The phase space is therefore the unit sphere (1). A canonical pair of coordinates on it is $(q, p_{\mathrm{can}}) = (\phi, J_z)$, or after dividing by j, (ϕ, Z). To check this, write $J_x = R\cos\phi$, $J_y = R\sin\phi$ with $R = \sqrt{j^2 - J_z^2}$ (so $\partial R/\partial J_z = -J_z/R$) and use the canonical bracket $\{A,B\} = \partial_\phi A\, \partial_{J_z} B - \partial_{J_z} A\, \partial_\phi B$:

$$\{J_x, J_y\} = (-R\sin\phi)\Big(-\frac{J_z}{R}\sin\phi\Big) - \Big(-\frac{J_z}{R}\cos\phi\Big)(R\cos\phi) = J_z(\sin^2\phi + \cos^2\phi) = J_z,$$

and in the same way $\{J_y, J_z\} = J_x$, $\{J_z, J_x\} = J_y$. These are the classical angular-momentum brackets, so (ϕ, J_z) is canonical.

Phase-space area equals geometric area. The Liouville (phase-space) area element in canonical coordinates is $d\phi\, dJ_z = j\, d\phi\, dZ$. The geometric area element of the unit sphere is $\sin\theta\, d\theta\, d\phi$. Since $Z = \cos\theta$, $dZ = -\sin\theta\, d\theta$, and so (orienting the Z integral from -1 to 1)

$$d\Omega \equiv \sin\theta\, d\theta\, d\phi = d\phi\, dZ, \qquad \int d\Omega = \int_{-\pi}^{\pi} d\phi \int_{-1}^{1} dZ = 4\pi. \tag{6}$$

Equal areas in the (ϕ, Z) plane are equal areas on the sphere (Archimedes' "hat-box" theorem). This is the argument behind `equal_area_grid` in `src/common/sphere_mod.f90`: a grid that is uniform in (ϕ, Z) has cells of equal area on the sphere (section 3.5). The classical map conserves this area (Liouville's theorem), so a cloud of classical points keeps its area while it is stretched and folded.

Why a quantum phase-space picture is needed. A quantum state of the top is a vector of $2j+1$ amplitudes ψ_m. The numbers $|\psi_m|^2$ give the probability distribution of J_z only, i.e. of one coordinate ($Z = m/j$); the information about ϕ sits in the relative phases of the ψ_m and is invisible in $|\psi_m|^2$. A quantum state also cannot be a point: by (2) the three components of $\boldsymbol{J}$ cannot all be sharp. What we want is a non-negative function on the sphere, normalised like a probability density, that (i) is concentrated near $\hat{\boldsymbol{n}}$ for a state "pointing" along $\hat{\boldsymbol{n}}$, (ii) is plotted on the same (ϕ, Z) axes as the classical phase portraits, and (iii) can be computed cheaply from the ψ_m. The Husimi function does all three. Its building block is the quantum state that is as close as possible to a point of the sphere: the spin coherent state.

2 Spin coherent states

2.1 Spin j as $2j$ spins $\frac{1}{2}$

Take $n = 2j$ spins $\frac{1}{2}$ (qubits) with Pauli matrices σ_i^a for qubit i; in the basis $(|\uparrow\rangle, |\downarrow\rangle)$

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \text{e.g. } \sigma^y|\uparrow\rangle = i|\downarrow\rangle,\ \sigma^y|\downarrow\rangle = -i|\uparrow\rangle. \quad (7)$$

The collective spin $\boldsymbol{S} = \frac{1}{2}\sum_{i=1}^{n} \boldsymbol{\sigma}_i$ satisfies (2), because operators of different qubits commute and $[\sigma_i^a/2, \sigma_i^b/2] = i\epsilon_{abc}\sigma_i^c/2$. A product basis state with u qubits up and $n-u$ down has $S_z = \frac{1}{2}(u - (n-u)) = u - \frac{n}{2}$. Define the *Dicke state* as the normalised equal-weight sum of all $\binom{n}{u}$ such product states,

$$|D_u\rangle = \binom{n}{u}^{-1/2} \sum_{u \text{ up}} |\cdots\rangle, \qquad S_z|D_u\rangle = \Big(u - \frac{n}{2}\Big)|D_u\rangle. \quad (8)$$

It is symmetric under every permutation of the qubits. We now show that $|D_u\rangle = |j, m\rangle$ with $j = n/2$ and $m = u - j$, including the phases of (3).

Step 1. The all-up state $|D_n\rangle = |\uparrow\uparrow\cdots\uparrow\rangle$ has $S_z = n/2 = j$, the largest possible value, and $S_+ = \sum_i \sigma_i^+$ (with $\sigma^+ = |\uparrow\rangle\langle\downarrow|$) annihilates it. So it is $|j, j\rangle$.

Step 2. $S_- = \sum_i \sigma_i^-$ with $\sigma^- = |\downarrow\rangle\langle\uparrow|$ turns one up qubit down. Acting on $|D_u\rangle$, every one of its $\binom{n}{u}$ terms produces u terms with $u-1$ up qubits. A given product state with $u-1$ up qubits is produced from each of the states obtained by turning one of its $n-u+1$ down qubits up, i.e. $n-u+1$ times. Therefore

$$S_-|D_u\rangle = \binom{n}{u}^{-1/2} (n-u+1) \sum_{u-1 \text{ up}} |\cdots\rangle = (n-u+1)\binom{n}{u}^{-1/2}\binom{n}{u-1}^{1/2}|D_{u-1}\rangle.$$

With $\binom{n}{u-1}/\binom{n}{u} = \frac{u!\,(n-u)!}{(u-1)!\,(n-u+1)!} = \frac{u}{n-u+1}$ the coefficient is $\sqrt{u\,(n-u+1)}$. Inserting $u = j+m$ and $n-u+1 = j-m+1$,

$$S_-|D_u\rangle = \sqrt{(j+m)(j-m+1)}\,|D_{u-1}\rangle = \sqrt{j(j+1) - m(m-1)}\,|D_{u-1}\rangle = a(m-1)\,|D_{u-1}\rangle,$$

which is exactly $J_-|j,m\rangle = a(m-1)|j,m-1\rangle$ of (3). Starting from Step 1 and applying S_- repeatedly, $|D_u\rangle = |j, u-j\rangle$ for all u. The $2j+1$ Dicke states span the *symmetric subspace* and in it $\boldsymbol{S}$ acts as the spin-j operator $\boldsymbol{J}$. In the rest of this section we write $\boldsymbol{J}$ for $\boldsymbol{S}$.

2.2 Rotating the spin operators

We will need how a rotation operator acts on $\boldsymbol{J}$. Let $F(\alpha) = e^{i\alpha J_z} J_x e^{-i\alpha J_z}$ and $G(\alpha) = e^{i\alpha J_z} J_y e^{-i\alpha J_z}$. Differentiating and using (2),

$$F' = e^{i\alpha J_z}\, i[J_z, J_x]\, e^{-i\alpha J_z} = e^{i\alpha J_z}\, i(iJ_y)\, e^{-i\alpha J_z} = -G, \qquad G' = e^{i\alpha J_z}\, i(-iJ_x)\, e^{-i\alpha J_z} = F,$$

with $F(0) = J_x$, $G(0) = J_y$. The solution is $F = J_x\cos\alpha - J_y\sin\alpha$, $G = J_x\sin\alpha + J_y\cos\alpha$ (check: $F' = -J_x\sin\alpha - J_y\cos\alpha = -G$, $G' = J_x\cos\alpha - J_y\sin\alpha = F$), while J_z commutes with $e^{i\alpha J_z}$. Comparing with (5), and repeating the same steps for J_y (where $i[J_y, J_z] = -J_x$ and $i[J_y, J_x] = J_z$),

$$e^{i\alpha J_z} J_a\, e^{-i\alpha J_z} = \sum_b R_z(\alpha)_{ab}\, J_b, \qquad e^{i\alpha J_y} J_a\, e^{-i\alpha J_y} = \sum_b R_y(\alpha)_{ab}\, J_b. \quad (9)$$

For $D(\theta, \phi) = e^{-i\phi J_z} e^{-i\theta J_y}$ we apply the two identities one after the other:

$$D^\dagger J_a D = e^{i\theta J_y}\big(e^{i\phi J_z} J_a e^{-i\phi J_z}\big)e^{-i\theta J_y} = \sum_b R_z(\phi)_{ab}\, e^{i\theta J_y} J_b e^{-i\theta J_y} = \sum_c \mathcal{R}_{ac}\, J_c, \quad (10)$$

with $\mathcal{R} = R_z(\phi)R_y(\theta)$. This matrix carries $\hat{\boldsymbol{z}}$ to $\hat{\boldsymbol{n}}$: $R_y(\theta)\hat{\boldsymbol{z}} = (\sin\theta, 0, \cos\theta)$ (third column of R_y) and $R_z(\phi)$ turns this into $(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta) = \hat{\boldsymbol{n}}$. For a fixed vector $\boldsymbol{e}$, (10) gives $D^\dagger(\boldsymbol{e}\cdot\boldsymbol{J})D = \sum_{a,c} e_a \mathcal{R}_{ac} J_c = (\mathcal{R}^\mathsf{T}\boldsymbol{e})\cdot\boldsymbol{J}$, where $^\mathsf{T}$ is the transpose and $\mathcal{R}^\mathsf{T} = \mathcal{R}^{-1}$.

2.3 Definition and amplitudes

The state $|j, j\rangle$ points along $+\hat{\boldsymbol{z}}$. Rotating it to $\hat{\boldsymbol{n}}(\theta,\phi)$ defines the spin coherent state

$$|\theta,\phi\rangle \equiv |\hat{\boldsymbol{n}}\rangle = D(\theta,\phi)\,|j,j\rangle = e^{-i\phi J_z}\, e^{-i\theta J_y}\,|j,j\rangle. \tag{11}$$

To find its amplitudes we use section 2.1. Since $J_y = \frac{1}{2}\sum_i \sigma_i^y$ is a sum of commuting terms, $e^{-i\theta J_y} = \prod_i e^{-i\theta\sigma_i^y/2}$, and the same for J_z. For one qubit, $(\sigma^y)^2 = 1$ splits the exponential series into even and odd powers,

$$e^{-i\theta\sigma^y/2} = \sum_r \frac{(-i\theta/2)^r}{r!}(\sigma^y)^r = \cos\frac{\theta}{2} - i\sin\frac{\theta}{2}\,\sigma^y,$$

$$e^{-i\theta\sigma^y/2}|\uparrow\rangle = \cos\frac{\theta}{2}|\uparrow\rangle - i\sin\frac{\theta}{2}\,(i|\downarrow\rangle) = \cos\frac{\theta}{2}|\uparrow\rangle + \sin\frac{\theta}{2}|\downarrow\rangle,$$

and $e^{-i\phi\sigma^z/2}$ multiplies $|\uparrow\rangle$ by $e^{-i\phi/2}$ and $|\downarrow\rangle$ by $e^{+i\phi/2}$. Hence each qubit ends up in the single-qubit coherent state

$$|\hat{\boldsymbol{n}}\rangle_1 = \alpha|\uparrow\rangle + \beta|\downarrow\rangle, \qquad \alpha = \cos\frac{\theta}{2}\,e^{-i\phi/2}, \quad \beta = \sin\frac{\theta}{2}\,e^{i\phi/2}, \tag{12}$$

and the spin coherent state is the product state $|\hat{\boldsymbol{n}}\rangle = |\hat{\boldsymbol{n}}\rangle_1^{\otimes 2j}$. Expanding the product, a product basis state with u qubits up gets the amplitude $\alpha^u\beta^{2j-u}$, which depends only on u. With $u = j+m$ and $2j-u = j-m$,

$$\alpha^{j+m}\beta^{j-m} = \cos^{j+m}\frac{\theta}{2}\,\sin^{j-m}\frac{\theta}{2}\,e^{-i\phi[(j+m)-(j-m)]/2} = \cos^{j+m}\frac{\theta}{2}\,\sin^{j-m}\frac{\theta}{2}\,e^{-im\phi}.$$

All $\binom{2j}{j+m}$ product states with $u = j+m$ have this amplitude, so their sum is $\binom{2j}{j+m}^{1/2}|j,m\rangle$ by (8). Using $\binom{2j}{j+m} = \binom{2j}{j-m}$,

$$\boxed{|\theta,\phi\rangle = \sum_{m=-j}^{j} c_m(\theta,\phi)\,|j,m\rangle, \qquad c_m(\theta,\phi) = \sqrt{\binom{2j}{j-m}}\,\cos^{j+m}\frac{\theta}{2}\,\sin^{j-m}\frac{\theta}{2}\,e^{-im\phi}.} \tag{13}$$

This is the formula of `coherent_state` in `src/quantum/spin_mod.f90` (and eq. [eq:coherent] of `docs/lecture_note/kicked_top_notes.tex`). With the phases of (12) no overall phase has to be dropped.

Normalisation. By the binomial theorem $(x+y)^n = \sum_{r=0}^n \binom{n}{r}x^{n-r}y^r$ with $r = j-m$, $n = 2j$, $x = \cos^2\frac{\theta}{2}$, $y = \sin^2\frac{\theta}{2}$:

$$\sum_m |c_m|^2 = \sum_{r=0}^{2j}\binom{2j}{r}\left(\cos^2\frac{\theta}{2}\right)^{2j-r}\left(\sin^2\frac{\theta}{2}\right)^r = \left(\cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}\right)^{2j} = 1. \tag{14}$$

Mean spin. In $|j,j\rangle$, $\langle J_z\rangle = j$ and $\langle J_x\rangle = \langle J_y\rangle = 0$ (J_x and J_y are combinations of $J_\pm$, which change m, so their diagonal elements vanish). With (10),

$$\langle\hat{\boldsymbol{n}}|J_a|\hat{\boldsymbol{n}}\rangle = \langle j,j|D^\dagger J_a D|j,j\rangle = \sum_c \mathcal{R}_{ac}\langle j,j|J_c|j,j\rangle = j\,\mathcal{R}_{az} = j\,\hat{n}_a, \quad \text{i.e.} \quad \langle\boldsymbol{J}\rangle = j\,\hat{\boldsymbol{n}}. \tag{15}$$

Transverse width. In $|j,j\rangle$, write $J_x = \frac{1}{2}(J_+ + J_-)$, so $J_x^2 = \frac{1}{4}(J_+^2 + J_-^2 + J_+J_- + J_-J_+)$. $J_\pm^2$ change m by ± 2 and have zero diagonal elements; $J_+|j,j\rangle = 0$; and $J_+J_-|j,j\rangle = J_+\, a(j-1)|j,j-1\rangle = a(j-1)^2|j,j\rangle = 2j\,|j,j\rangle$. Hence $\langle J_x^2\rangle = \frac{2j}{4} = \frac{j}{2}$, and in the same way $\langle J_y^2\rangle = \frac{j}{2}$ (check: $\langle J_x^2 + J_y^2\rangle = \langle \boldsymbol{J}^2 - J_z^2\rangle = j(j+1) - j^2 = j$). Take the unit vectors $\boldsymbol{e}_1 = \mathcal{R}\hat{\boldsymbol{x}}$ and $\boldsymbol{e}_2 = \mathcal{R}\hat{\boldsymbol{y}}$ perpendicular to $\hat{\boldsymbol{n}}$. By the last line of section 2.2, $D^\dagger(\boldsymbol{e}_1\cdot\boldsymbol{J})D = (\mathcal{R}^\mathsf{T}\mathcal{R}\hat{\boldsymbol{x}})\cdot\boldsymbol{J} = J_x$, so in the coherent state

$$\langle \boldsymbol{e}_1\cdot\boldsymbol{J}\rangle = 0, \qquad \Delta(\boldsymbol{e}_1\cdot\boldsymbol{J}) = \sqrt{\langle J_x^2\rangle_{jj}} = \sqrt{j/2}, \qquad \Delta(\boldsymbol{e}_1\cdot\boldsymbol{s}) = \frac{\sqrt{j/2}}{j} = \frac{1}{\sqrt{2j}}, \tag{16}$$

and the same for $\boldsymbol{e}_2$. The scaled spin $\boldsymbol{s} = \boldsymbol{J}/j$ is spread over a patch of radius $\sim 1/\sqrt{2j}$ around $\hat{\boldsymbol{n}}$. The product $\Delta J_x \Delta J_y = j/2 = |\langle J_z\rangle|/2$ in $|j,j\rangle$ saturates the uncertainty relation $\Delta J_x \Delta J_y \geq \frac{1}{2}|\langle [J_x, J_y]\rangle|$: a coherent state is a minimum-uncertainty state.

Overlap of two coherent states. Both $|\hat{\boldsymbol{n}}\rangle$ and $|\hat{\boldsymbol{n}}'\rangle$ are products of $2j$ identical qubit states, and the inner product of product states is the product of the inner products: $\langle\hat{\boldsymbol{n}}|\hat{\boldsymbol{n}}'\rangle = (\langle\hat{\boldsymbol{n}}|\hat{\boldsymbol{n}}'\rangle_1)^{2j}$. For one qubit in (12), using (7),

$$\langle\sigma^z\rangle = |\alpha|^2 - |\beta|^2 = \cos\theta, \quad \langle\sigma^x\rangle = \alpha^*\beta + \beta^*\alpha = 2\,\mathrm{Re}(\alpha^*\beta), \quad \langle\sigma^y\rangle = -i\alpha^*\beta + i\beta^*\alpha = 2\,\mathrm{Im}(\alpha^*\beta),$$

and $\alpha^*\beta = \cos\frac{\theta}{2}\sin\frac{\theta}{2}\,e^{i\phi} = \frac{1}{2}\sin\theta\, e^{i\phi}$, so $\langle\boldsymbol{\sigma}\rangle = \hat{\boldsymbol{n}}$. Any 2×2 Hermitian matrix ρ with $\mathrm{Tr}\,\rho = 1$ can be written $\rho = \frac{1}{2}(1 + \boldsymbol{r}\cdot\boldsymbol{\sigma})$, and $\mathrm{Tr}(\sigma^a\sigma^b) = 2\delta_{ab}$, $\mathrm{Tr}\,\sigma^a = 0$ give $r_a = \mathrm{Tr}(\rho\sigma^a)$. For $\rho = |\hat{\boldsymbol{n}}\rangle_1\langle\hat{\boldsymbol{n}}|$ this is $r_a = \langle\sigma^a\rangle = \hat{n}_a$. Then

$$|\langle\hat{\boldsymbol{n}}|\hat{\boldsymbol{n}}'\rangle_1|^2 = \mathrm{Tr}(\rho\rho') = \frac{1}{4}\,\mathrm{Tr}\left[(1+\hat{\boldsymbol{n}}\cdot\boldsymbol{\sigma})(1+\hat{\boldsymbol{n}}'\cdot\boldsymbol{\sigma})\right] = \frac{1}{4}\big(2 + 2\,\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{n}}'\big) = \frac{1+\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{n}}'}{2},$$

and raising to the power $2j$,

$$\boxed{|\langle\hat{\boldsymbol{n}}|\hat{\boldsymbol{n}}'\rangle|^2 = \Big(\frac{1+\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{n}}'}{2}\Big)^{2j} = \cos^{4j}\frac{\gamma}{2},} \qquad \cos\gamma = \hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{n}}', \tag{17}$$

with γ the angle between the two directions, using $\frac{1+\cos\gamma}{2} = \cos^2\frac{\gamma}{2}$. Check with (13): for $\hat{\boldsymbol{n}} = \hat{\boldsymbol{z}}$, $|\hat{\boldsymbol{n}}\rangle = |j,j\rangle$ and $|\langle j,j|\hat{\boldsymbol{n}}'\rangle|^2 = |c_j(\theta',\phi')|^2 = \cos^{4j}\frac{\theta'}{2}$, the same as (17) with $\gamma = \theta'$. Coherent states are not orthogonal; their overlap falls off over an angle $\gamma \sim 1/\sqrt{j}$ (section 3.4).

3 The Husimi function of the kicked top

3.1 Definition

The Husimi function of a normalised state $|\psi\rangle$ is the probability of finding it in the coherent state $|\theta,\phi\rangle$:

$$\boxed{Q(\theta,\phi) = |\langle\theta,\phi|\psi\rangle|^2 = \Big|\sum_m c_m^*(\theta,\phi)\,\psi_m\Big|^2 = \Big|\sum_m \sqrt{\tbinom{2j}{j-m}}\cos^{j+m}\tfrac{\theta}{2}\sin^{j-m}\tfrac{\theta}{2}\,e^{im\phi}\,\psi_m\Big|^2.} \tag{18}$$

By the Cauchy–Schwarz inequality $|\langle\hat{\boldsymbol{n}}|\psi\rangle|^2 \leq \langle\hat{\boldsymbol{n}}|\hat{\boldsymbol{n}}\rangle\langle\psi|\psi\rangle = 1$, so $0 \leq Q \leq 1$. For the kicked top, Q_n is (18) for $|\psi_n\rangle = U^n|\psi_0\rangle$ of (4); the quantum program evaluates it at the kicks listed in `husimi_kicks`.

3.2 Resolution of the identity

Theorem.

$$A \equiv \frac{2j+1}{4\pi} \int |\theta,\phi\rangle\langle\theta,\phi|\, d\Omega = 1, \qquad d\Omega = d\phi\, dZ. \tag{19}$$

Proof. (a) A commutes with J_z. From (11), $e^{-i\alpha J_z}|\theta,\phi\rangle = |\theta,\phi+\alpha\rangle$. Hence $e^{-i\alpha J_z} A e^{i\alpha J_z} = \frac{2j+1}{4\pi}\int |\theta,\phi+\alpha\rangle\langle\theta,\phi+\alpha|\, d\phi\, dZ = A$, because the integrand is 2π-periodic in ϕ (for half-integer j, $|\theta,\phi+2\pi\rangle = -|\theta,\phi\rangle$, and the sign cancels in $|\cdot\rangle\langle\cdot|$). Taking $d/d\alpha$ at $\alpha = 0$ gives $-i[J_z, A] = 0$.

(b) A commutes with J_y. First, $|\hat{\boldsymbol{n}}\rangle$ is the eigenvector of $\hat{\boldsymbol{n}}\cdot\boldsymbol{J}$ with the largest eigenvalue j: by section 2.2, $D^\dagger(\hat{\boldsymbol{n}}\cdot\boldsymbol{J})D = (\mathcal{R}^\mathsf{T}\hat{\boldsymbol{n}})\cdot\boldsymbol{J} = \hat{\boldsymbol{z}}\cdot\boldsymbol{J} = J_z$, so $(\hat{\boldsymbol{n}}\cdot\boldsymbol{J})D|j,j\rangle = DJ_z|j,j\rangle = j\,D|j,j\rangle$. This eigenvalue is non-degenerate, since $\hat{\boldsymbol{n}}\cdot\boldsymbol{J} = DJ_zD^\dagger$ has the same eigenvalues $j,\dots,-j$ as J_z. Now let $V = e^{-i\beta J_y}$ and $\hat{\boldsymbol{n}}' = R_y(\beta)\hat{\boldsymbol{n}}$. By (9), $V^\dagger(\hat{\boldsymbol{n}}'\cdot\boldsymbol{J})V = (R_y(\beta)^\mathsf{T}\hat{\boldsymbol{n}}')\cdot\boldsymbol{J} = \hat{\boldsymbol{n}}\cdot\boldsymbol{J}$, so $(\hat{\boldsymbol{n}}'\cdot\boldsymbol{J})V|\hat{\boldsymbol{n}}\rangle = V(\hat{\boldsymbol{n}}\cdot\boldsymbol{J})|\hat{\boldsymbol{n}}\rangle = j\,V|\hat{\boldsymbol{n}}\rangle$: $V|\hat{\boldsymbol{n}}\rangle$ is the eigenvector of $\hat{\boldsymbol{n}}'\cdot\boldsymbol{J}$ with eigenvalue j, i.e. $|\hat{\boldsymbol{n}}'\rangle$ up to a phase, which cancels in $V|\hat{\boldsymbol{n}}\rangle\langle\hat{\boldsymbol{n}}|V^\dagger = |\hat{\boldsymbol{n}}'\rangle\langle\hat{\boldsymbol{n}}'|$. A rotation maps the sphere onto itself and preserves areas, so the change of variables $\hat{\boldsymbol{n}} \to \hat{\boldsymbol{n}}'$ leaves $d\Omega$ unchanged and $VAV^\dagger = A$. Differentiating in β at 0 gives $[J_y, A] = 0$.

(c) Schur's lemma for spin j. From (a) and (b), A also commutes with $J_x = -i[J_y, J_z]$ and hence with J_+. Since $J_zA|j,m\rangle = AJ_z|j,m\rangle = m\,A|j,m\rangle$ and the eigenvalue m of J_z is non-degenerate, $A|j,m\rangle = \lambda_m|j,m\rangle$. Then, for $m < j$,

$$AJ_+|j,m\rangle = a(m)\lambda_{m+1}|j,m+1\rangle \quad \text{and} \quad J_+A|j,m\rangle = \lambda_m a(m)|j,m+1\rangle,$$

and $a(m) \neq 0$ gives $\lambda_{m+1} = \lambda_m$. All λ_m are equal: $A = \lambda\,1$. (This is Schur's lemma for the irreducible spin-j representation: an operator that commutes with all rotations is a multiple of the identity.)

(d) The constant. Take the trace, using $\mathrm{Tr}(|\hat{\boldsymbol{n}}\rangle\langle\hat{\boldsymbol{n}}|) = \langle\hat{\boldsymbol{n}}|\hat{\boldsymbol{n}}\rangle = 1$ and (6): $\mathrm{Tr}\,A = \frac{2j+1}{4\pi}\int d\Omega = 2j+1$. Since $\mathrm{Tr}(\lambda 1) = \lambda(2j+1)$, $\lambda = 1$. □

Direct check. The matrix elements of A can also be computed directly, which confirms the theorem. With (13), $\langle j,m|A|j,m'\rangle = \frac{2j+1}{4\pi}\int c_m c^*_{m'}\, d\phi\, dZ$. The ϕ integral is $\int_{-\pi}^{\pi} e^{-i(m-m')\phi}d\phi = 2\pi\,\delta_{mm'}$ ($m - m'$ is an integer). For $m = m'$ substitute $x = \cos^2\frac{\theta}{2} = \frac{1+Z}{2}$, so $\sin^2\frac{\theta}{2} = 1-x$ and $dZ = 2\,dx$:

$$\int_{-1}^{1}|c_m|^2 dZ = \binom{2j}{j-m}\int_0^1 x^{j+m}(1-x)^{j-m}\,2\,dx = \frac{2\,(2j)!}{(j-m)!\,(j+m)!}\,\frac{(j+m)!\,(j-m)!}{(2j+1)!} = \frac{2}{2j+1},$$

where we used $\int_0^1 x^a(1-x)^b dx = \frac{a!\,b!}{(a+b+1)!}$ (integrate by parts b times: each step gives $\int_0^1 x^a(1-x)^b dx = \frac{b}{a+1}\int_0^1 x^{a+1}(1-x)^{b-1}dx$, and at the end $\int_0^1 x^{a+b}dx = \frac{1}{a+b+1}$). Altogether $\langle j,m|A|j,m'\rangle = \frac{2j+1}{4\pi}\cdot 2\pi\cdot\frac{2}{2j+1}\,\delta_{mm'} = \delta_{mm'}$.

3.3 Normalisation of Q

Take $\langle\psi|\cdot|\psi\rangle$ of (19): $\langle\psi|\theta,\phi\rangle\langle\theta,\phi|\psi\rangle = Q$, so

$$\boxed{\frac{2j+1}{4\pi}\int Q(\theta,\phi)\,d\phi\,dZ = \langle\psi|\psi\rangle = 1.} \tag{20}$$

So $\frac{2j+1}{4\pi}Q$ is a probability density on the sphere with respect to $d\Omega$. A state "occupies" the area $4\pi/(2j+1)$ (a *Planck cell*): the $2j+1$ orthogonal states $|j,m\rangle$ tile the area 4π.

3.4 Worked examples

(a) The state $|j, j\rangle$. $\psi_m = \delta_{mj}$, so by (18) and (13) with $m = j$ ($\binom{2j}{0} = 1$),

$$Q(\theta, \phi) = \cos^{4j}\frac{\theta}{2} = \left(\frac{1+Z}{2}\right)^{2j}. \tag{21}$$

It does not depend on ϕ, equals 1 at the north pole $Z = 1$ and decays towards the south pole. Normalisation: $\frac{2j+1}{4\pi} \cdot 2\pi \int_{-1}^{1} \left(\frac{1+Z}{2}\right)^{2j} dZ = \frac{2j+1}{2} \cdot \frac{2}{2j+1} = 1$, as (20) requires (substitute $x = (1+Z)/2$: $\int_{-1}^{1} (\frac{1+Z}{2})^{2j} dZ = 2\int_0^1 x^{2j} dx = \frac{2}{2j+1}$).

(b) A coherent state. For $|\psi\rangle = |\hat{\boldsymbol{n}}_0\rangle$, (18) is the overlap (17):

$$Q(\hat{\boldsymbol{n}}) = \left(\frac{1 + \hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}}_0}{2}\right)^{2j} = \cos^{4j}\frac{\gamma}{2}, \qquad \cos\gamma = \hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}}_0. \tag{22}$$

For small γ, $\ln\cos x = -\frac{x^2}{2} - \frac{x^4}{12} - \ldots$, so $\ln Q = 4j \ln\cos\frac{\gamma}{2} \approx -4j\,\frac{\gamma^2}{8} = -\frac{j\gamma^2}{2}$ and

$$Q \approx e^{-\gamma^2/(2\sigma^2)}, \qquad \sigma = \frac{1}{\sqrt{j}}. \tag{23}$$

The coherent state is a Gaussian blob of angular width $1/\sqrt{j}$ around $\hat{\boldsymbol{n}}_0$. The neglected term, $4j(\gamma/2)^4/12 = j\gamma^4/48$, is of order $1/(48j)$ for $\gamma \sim \sigma$. Example: $j = 50$ gives $\sigma = 0.14$, and Q drops to $1/2$ at $\gamma = \sqrt{2\ln 2/j} = 0.17$. (The width $1/\sqrt{j}$ of Q is $\sqrt{2}$ times the width $1/\sqrt{2j}$ of the state itself in (16), because Q is the overlap of two blobs.) Case (a) is (22) with $\hat{\boldsymbol{n}}_0 = \hat{\boldsymbol{z}}$.

(c) A Dicke state $|j, m_0\rangle$. $\psi_m = \delta_{mm_0}$ and, with $x = \frac{1+Z}{2}$, $Q = |c_{m_0}|^2 = \binom{2j}{j-m_0} x^{j+m_0}(1-x)^{j-m_0}$, independent of ϕ: a ring around the z axis. Its maximum follows from $\frac{d}{dx}\ln Q = \frac{j+m_0}{x} - \frac{j-m_0}{1-x} = 0$, i.e. $x = \frac{j+m_0}{2j}$, or $Z = 2x - 1 = m_0/j$: the ring sits at the classical height $J_z = m_0$.

3.5 Numerical evaluation

The equal-area grid. `equal_area_grid` (`sphere_mod`) builds $n_\theta \times n_\phi$ cell centres, uniform in Z and ϕ,

$$Z_i = -1 + \left(i - \frac{1}{2}\right)\frac{2}{n_\theta}, \quad \theta_i = \arccos Z_i, \qquad \phi_l = -\pi + \left(l - \frac{1}{2}\right)\frac{2\pi}{n_\phi}, \tag{24}$$

$i = 1, \ldots, n_\theta$ (outer loop), $l = 1, \ldots, n_\phi$ (inner loop). Each cell is a rectangle $\Delta Z \times \Delta\phi = \frac{2}{n_\theta} \times \frac{2\pi}{n_\phi}$ in the (ϕ, Z) plane, so by (6) every cell has the same area on the sphere,

$$\Delta\Omega = \frac{4\pi}{n_\theta n_\phi} = \frac{4\pi}{N}, \qquad N = n_\theta n_\phi. \tag{25}$$

A grid uniform in θ would instead crowd its points near the poles.

Q as a complex dot product. `husimi` (`src/quantum/husimi_mod.f90`) evaluates (18) at every grid point. Fortran's `dot_product(a, b)` for complex arrays is $\sum_i a_i^* b_i$, so with $a =$ the amplitudes (13) and $b = \psi$ it is exactly $\langle\theta, \phi|\psi\rangle = \sum_m c_m^* \psi_m$; the loop body is `q(i)= abs( dot_product(coherent_state(j, size(psi), theta(i), phi(i)), psi))**2`. The grid points are independent and are distributed over threads with OpenMP. The cost is $(2j+1)$ operations per point.

The discrete normalisation check. Replacing the integral in (20) by the sum over cell centres (midpoint rule) with (25),

$$\frac{2j+1}{4\pi}\int Q\,d\Omega \approx \frac{2j+1}{4\pi}\sum_{i,l} Q(\theta_i,\phi_l)\,\frac{4\pi}{N} = (2j+1)\,\frac{1}{N}\sum_{i,l} Q(\theta_i,\phi_l) = (2j+1)\cdot\mathrm{mean}(Q), \quad (26)$$

which is `husimi_norm`: `norm = (2*j+1)*sum(q)/size(q)`. It should be 1. How accurate is it? Expanding (18), $Q=\sum_{m,m'} c_m^* c_{m'}\psi_m\psi_{m'}^*$ contains ϕ only through $e^{i(m-m')\phi}$ with $K=m-m'\in[-2j,2j]$. On the grid (24), with $\Delta=2\pi/n_\phi$ and the geometric sum,

$$\sum_{l=1}^{n_\phi} e^{iK\phi_l} = e^{iK(-\pi+\Delta/2)}\sum_{l=0}^{n_\phi-1}\left(e^{iK\Delta}\right)^l = e^{iK(-\pi+\Delta/2)}\,\frac{1-e^{2\pi iK}}{1-e^{iK\Delta}} = 0 \quad \text{unless } n_\phi \mid K.$$

($n_\phi \mid K$: K is a multiple of n_ϕ.) So for $n_\phi \geq 2j+1$ only $K=0$ survives and the ϕ sum is exact. What remains is $\sum_m |\psi_m|^2|c_m|^2$, a polynomial of degree $2j$ in Z, integrated with the midpoint rule, whose error decreases as $1/n_\theta^2$. For example, evaluating (26) for a coherent state with $j=7/2$, $\theta_0=1.1$, $\phi_0=0.7$ gives 1.00103 on a 20×40 grid and 1.00004 on a 100×200 grid (the default of `config_quantum.nml`); the ratio of the errors is $\approx 25=(100/20)^2$.

The log-binomial trick. For large j the factors in (13) are individually out of the range of double precision (largest number $\approx 1.8\times10^{308}$): $\binom{2j}{j}\approx 4^j/\sqrt{\pi j}$ is about 10^{360} for $j=600$, while $\cos^{j+m}\frac{\theta}{2}\sin^{j-m}\frac{\theta}{2}$ is correspondingly tiny; their product $|c_m|\leq 1$ is harmless. `coherent_state` therefore works with logarithms. It builds $\ln\binom{2j}{r}$, $r=j-m=i-1$, by the recursion $\binom{2j}{r}=\binom{2j}{r-1}\frac{2j-r+1}{r}$,

$$\ln\binom{2j}{r} = \ln\binom{2j}{r-1} + \ln\frac{2j-(r-1)}{r}, \qquad \ln\binom{2j}{0}=0, \quad (27)$$

adds $(j+m)\ln\cos\frac{\theta}{2}+(j-m)\ln\sin\frac{\theta}{2}$, and exponentiates only the sum:

```
log_binom = 0.0_wp ! log binom(2j, r), r = j - m = i - 1
do i = 1, n
  r = i - 1
  if (r > 0) log_binom = log_binom + log((2.0_wp*j - real(r - 1, wp))/real(r, wp))
  log_amp = 0.5_wp*log_binom + log_power(cos(0.5_wp*theta), j + m(i)) &
          + log_power(sin(0.5_wp*theta), j - m(i))
  c(i) = exp(log_amp)*cmplx(cos(m(i)*phi), -sin(m(i)*phi), wp)
end do
```

`log_power(x, e)` returns $e\ln x$, with $0^0=1$ (returns 0) and, for $x=0$, $e>0$, a huge negative number so that `exp` gives 0 (this happens at the poles, $\theta=0$ or π). The phase $e^{-im\phi}$ is `cmplx(cos(m*phi), -sin(m*phi))`.

4 The Husimi function of the qubit chain

4.1 Chain, collective spin and Dicke states

The chain has n_q qubits. A computational basis state $|b\rangle$ is labelled by an integer $b=0,\ldots,2^{n_q}-1$: bit $i-1$ of b is 1 if qubit i is up ($\sigma_i^z=+1$) and 0 if it is down. The number of up qubits is $u=\mathrm{popcnt}(b)$, the number of set bits. The state is the vector $\psi(b)=\langle b|\psi\rangle$ of length 2^{n_q}; no $2^{n_q}\times 2^{n_q}$ matrix is ever stored. The collective spin is

$$\boldsymbol{S}=\frac{1}{2}\sum_{i=1}^{n_q}\boldsymbol{\sigma}_i, \qquad S_z|b\rangle=\left(u-\frac{n_q}{2}\right)|b\rangle. \quad (28)$$

The chain is driven by (4) with $\boldsymbol{J} \to \boldsymbol{S}$, where S_z^2 is realised as the all-to-all Ising coupling $H_{zz} = \sum_{i,l} V_{il}\sigma_i^z\sigma_l^z$ with $V_{il} = \frac{1}{4}$ ($i = l$ included), so that $H_{zz} = S_z^2$.

By section 2.1 with $n = n_q$, the symmetric subspace is spanned by the Dicke states

$$|j,m\rangle = \binom{n_q}{u}^{-1/2} \sum_{b:\ \mathrm{popcnt}(b)=u} |b\rangle, \qquad j = \frac{n_q}{2}, \quad u = j+m, \tag{29}$$

and in this subspace $\boldsymbol{S}$ is the spin-j operator $\boldsymbol{J}$ of the kicked top. Each sum contains $\binom{n_q}{u}$ terms, which is why the prefactor normalises the state.

4.2 Projection onto the Dicke states: `dicke_amplitudes`

The amplitude of ψ on $|j,m\rangle$ follows from (29) and $\langle b|\psi\rangle = \psi(b)$:

$$\boxed{c_m \equiv \langle j,m|\psi\rangle = \binom{n_q}{u}^{-1/2} \sum_{b:\ \mathrm{popcnt}(b)=u} \psi(b), \qquad u = j+m.} \tag{30}$$

`dicke_amplitudes` (`src/chain/chain_state_mod.f90`) computes it in one pass over the 2^{n_q} amplitudes, accumulating $a(u) = \sum_{\mathrm{popcnt}(b)=u} \psi(b)$, then rescales:

```
a = (0.0_wp, 0.0_wp)
do b = 0, size(psi) - 1
  a(popcnt(b)) = a(popcnt(b)) + psi(b)
end do
do u = 0, n_q
  log_binom = log_gamma(real(n_q + 1, wp)) - log_gamma(real(u + 1, wp)) - log_gamma(
      real(n_q - u + 1, wp))
  c(n_q - u + 1) = a(u)*exp(-0.5_wp*log_binom)
end do
```

The binomial is evaluated with the log-Gamma function: $\Gamma(r+1) = r!$ for integer r, so

$$\ln\binom{n_q}{u} = \ln\Gamma(n_q+1) - \ln\Gamma(u+1) - \ln\Gamma(n_q-u+1), \qquad \binom{n_q}{u}^{-1/2} = \exp\left[-\frac{1}{2}\ln\binom{n_q}{u}\right], \tag{31}$$

which never forms the large factorials themselves. The result is stored at index $i = n_q - u + 1$, i.e. $m = j - (i-1) = j - n_q + u = u - j$, the ordering of `spin_mod`, so every spin-j routine can be applied to it.

Let $P = \sum_m |j,m\rangle\langle j,m|$ be the projector onto the symmetric subspace. Then $P|\psi\rangle = \sum_m c_m|j,m\rangle$ and

$$w \equiv \langle\psi|P|\psi\rangle = \sum_m |c_m|^2 \le 1, \tag{32}$$

with $w = 1$ exactly when ψ lies in the symmetric subspace.

4.3 Product coherent states are spin coherent states

The initial state of the chain is `product_coherent_state`: every qubit in the state (12) (`a_up` $= \alpha$, `a_dn` $= \beta$), and `psi(b)= amp(popcnt(b))` with `amp(u)= a_up**u*a_dn**(n_q - u)`. So $\psi(b) = \alpha^u\beta^{n_q-u}$, which depends on b only through u. Insert this into (30): the sum has $\binom{n_q}{u}$ equal terms, so

$$c_m = \binom{n_q}{u}^{-1/2}\binom{n_q}{u}\alpha^u\beta^{n_q-u} = \sqrt{\binom{n_q}{u}}\,\alpha^{j+m}\beta^{j-m} = \sqrt{\binom{2j}{j-m}}\,\cos^{j+m}\frac{\theta}{2}\,\sin^{j-m}\frac{\theta}{2}\,e^{-im\phi},$$

using $n_q = 2j$, $\binom{2j}{j+m} = \binom{2j}{j-m}$ and the algebra of section 2.3. This is exactly (13):

$$\texttt{dicke_amplitudes}\big(\texttt{product_coherent_state}(n_q, \theta, \phi)\big) = \texttt{coherent_state}(n_q/2,\, n_q+1,\, \theta,\, \phi), \tag{33}$$

with no phase difference, and $w = 1$ by (14). Numerically the two sides agree to machine precision ($\sim 10^{-16}$). This is no coincidence: section 2.3 constructed the spin coherent state precisely as this product state.

4.4 Husimi function of the chain

Define the Husimi function of a chain state as the probability of finding it in the product coherent state, $Q(\theta,\phi) = |(\langle\hat{\boldsymbol{n}}|_1)^{\otimes n_q}|\psi\rangle|^2$. By (33) the product coherent state lies in the symmetric subspace, $(\langle\hat{\boldsymbol{n}}|_1)^{\otimes n_q} = \langle\hat{\boldsymbol{n}}| = \langle\hat{\boldsymbol{n}}|P$, and therefore

$$Q(\theta,\phi) = |\langle\hat{\boldsymbol{n}}|P|\psi\rangle|^2 = \Big|\sum_m c_m^*(\theta,\phi)\,\langle j,m|\psi\rangle\Big|^2, \tag{34}$$

which is (18) with ψ_m replaced by the Dicke amplitudes (30). The code (`run_chain` in `src/combined/chain_run_mod.f90`) does exactly this:

```
call husimi(dicke_amplitudes(psi, cfg%n_q), j, hus_theta, hus_phi, q(:, l))
```

The 2^{n_q}-dimensional state is reduced once to n_q+1 numbers (cost 2^{n_q}), and each grid point then costs only n_q+1 operations, the same as for the spin-j top.

5 Symmetry caveat: leaving the symmetric subspace

The dynamics keeps the symmetric subspace. The kick $e^{-ipS_y} = \prod_i e^{-ip\sigma_i^y/2}$ and the coupling $e^{-i\frac{k}{2j}H_{zz}}$ with uniform $V_{il} = \frac{1}{4}$ are unchanged when qubits are permuted. The symmetric subspace is the span of $|j,j\rangle$ and the states $S_-^r|j,j\rangle$, and these operators are functions of $\boldsymbol{S}$, which maps it into itself; so $UP = PUP$, and since $U^\dagger$ is also a function of $\boldsymbol{S}$, $PU = PUP$ too. Hence $[U,P] = 0$, and $w(t) = \langle\psi(t)|P|\psi(t)\rangle$ is conserved. The same holds for the backward step (time reversal with $k/(2j) \to 2\pi - k/(2j)$, $p \to -p$). Starting from a product coherent state, $w = 1$ on the whole forward leg.

A single-qubit perturbation leaks weight. With `butterfly_effect`, `rotate_qubit` applies $e^{-i\vartheta\sigma^a_{i_{\rm flip}}/2}$ at t_r: a rotation of qubit $i_{\rm flip}$ about $a =$ `flip_axis` by $\vartheta =$ `flip_angle` (default π, the flip $\sigma^a_{i_{\rm flip}}$ up to the phase $-i$). It acts on one qubit only and does not commute with permutations, so the state acquires a component $(1-P)|\psi\rangle$ outside the symmetric subspace. That component has $\boldsymbol{S}^2 < j(j+1)$ and lives in sectors of smaller total spin; it has no spin-j Dicke amplitudes. Afterwards w is again conserved, so on the whole backward leg

$$|\psi\rangle = \underbrace{P|\psi\rangle}_{\text{weight } w} + \underbrace{(1-P)|\psi\rangle}_{\text{weight } 1-w}. \tag{35}$$

What the plotted Q shows. By (34) Q only sees $P|\psi\rangle$: the plotted Husimi function is that of the symmetric component alone. Its normalisation follows from (19), which in the chain reads $\frac{2j+1}{4\pi}\int|\hat{\boldsymbol{n}}\rangle\langle\hat{\boldsymbol{n}}|\,d\Omega = P$ (the identity of the symmetric subspace):

$$\frac{2j+1}{4\pi}\int Q\,d\Omega = \langle\psi|P|\psi\rangle = w = 1 - (\text{leaked weight}). \tag{36}$$

So `husimi_norm` < 1 measures the weight that has left the symmetric subspace. `run_chain` also computes this weight directly from (32), without the grid error of (26):

```
sym_dev = max(sym_dev, abs(1.0_wp - sum(abs(dicke_amplitudes(psi, cfg%n_q))**2)))
```

at the five Husimi snapshots $t = 0, t_r/2, t_r, 3t_r/2, 2t_r$ (integer division; the snapshot at t_r is taken before the rotation). It is written to `summary.dat` as `max(1-symmetric_weight)`, next to `husimi_norm(0)` ... `husimi_norm(2t_r)`. Without a flip `sym_dev` ≈ 0; with a flip `sym_dev` $= 1 - w$ and `husimi_norm(2t_r)` $\approx 1 -$ `sym_dev`.

Worked example. Let the state at t_r be the all-up state $|j, j\rangle = |b = 2^{n_q} - 1\rangle$ (close to what happens in the regular regime with z polarisation) and flip qubit 1 about x ($\vartheta = \pi$). The result is the single basis state with qubit 1 down, $u = n_q - 1$. By (30) its only non-zero Dicke amplitude is $c_{j-1} = \binom{n_q}{n_q-1}^{-1/2} = 1/\sqrt{n_q}$, so $w = 1/n_q$ (for $n_q = 10$, $w = 0.1$): 90% of the state has left the symmetric subspace. The plotted $Q = \frac{1}{n_q} Q_{|j,j-1\rangle}$ is, by example (c) of section 3.4, a ring at $Z = (j-1)/j = 1 - 2/n_q$ with $\frac{2j+1}{4\pi} \int Q = 0.1$. A z flip instead, $\sigma_1^z |\uparrow \cdots \uparrow\rangle = |\uparrow \cdots \uparrow\rangle$, leaves the state symmetric ($w = 1$): whether weight leaks depends on the state and on the axis. For general ϑ, $e^{-i\vartheta\sigma_1^x/2} = \cos\frac{\vartheta}{2} - i \sin\frac{\vartheta}{2}\, \sigma_1^x$ gives $\cos\frac{\vartheta}{2}|\uparrow \cdots \uparrow\rangle - i \sin\frac{\vartheta}{2}|\downarrow\uparrow \cdots \uparrow\rangle$, so the leaked weight is $1 - w = (1 - \frac{1}{n_q}) \sin^2\frac{\vartheta}{2}$.

6 Plotting

The data file. `write_husimi` writes `husimi.dat`: in `src/quantum/quantum_io_mod.f90` the columns are `n phi Z X Y Q` with `n` the kick, in `src/combined/combined_io_mod.f90` they are `t phi Z X Y Q` with `t` $\in \{0, t_r/2, t_r, 3t_r/2, 2t_r\}$. One block of $N = n_\theta n_\phi$ lines is written per snapshot, in the order of `equal_area_grid` (Z outer, ϕ inner), blocks separated by a blank line. Z, X and Y come from `spherical_to_cartesian` (1). Fortran does all the computation; Python only reads and draws.

Rendering. `husimi()` in `docs/three_regime/figures/plot_three_regimes.py` (and `quantum_husimi()` in `docs/lecture_note/figures/plot_figures.py`) does

```
d = load(i, "chain/husimi.dat", data) # np.loadtxt skips the blank lines
b = d[d[:, 0] == t] # the block of snapshot t
shape = (-1, n_phi) # n_phi = number of distinct phi in a block
ax.pcolormesh(b[:, 1].reshape(shape), b[:, 2].reshape(shape), b[:, 5].reshape(shape),
              shading="nearest", cmap="viridis", rasterized=True)
```

Because ϕ runs fastest, `reshape(-1, n_phi)` gives an $n_\theta \times n_\phi$ array whose row i is the line $Z = Z_i$. With `shading="nearest"` every value is drawn as a rectangle centred on its grid point: these are exactly the equal-area cells (25). `phase_axes` sets $\phi \in [-\pi, \pi]$ and $Z \in [-1, 1]$, the same axes as the classical Poincaré sections.

Why (ϕ, Z) axes.

1. *Areas are phase-space areas.* By (6) a region of the plot has the same area as the corresponding region of the sphere (the same Liouville measure); in (ϕ, θ) axes the poles would be stretched.

2. *Classical comparison.* (ϕ, Z) is the canonical pair of section 1. Classical phase portraits are plotted in the same coordinates, so a Husimi plot can be laid directly over a portrait: a blob that sits on a regular island, or that has spread over the chaotic sea, is seen at once.

3. *Normalisation by eye.* Every pixel has the same weight, so (26) is $(2j + 1)$ times the plain mean of the plotted values; the code checks it as `husimi_norm`.